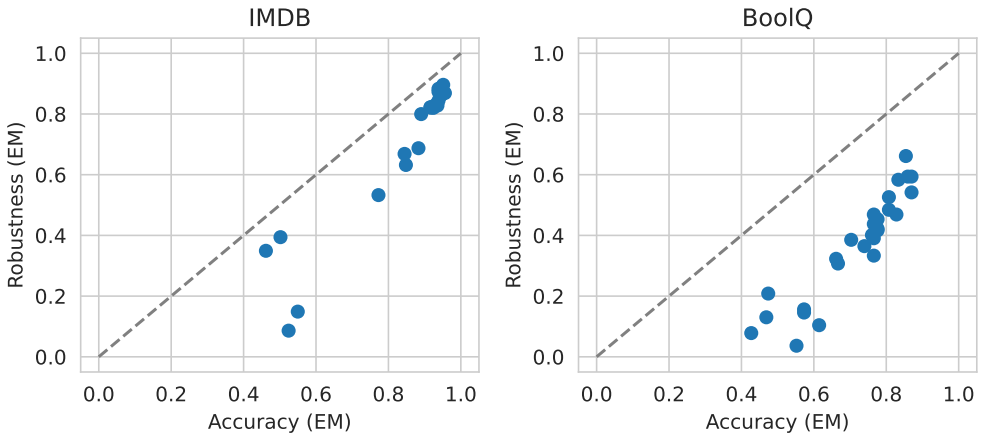


Figure 35: **Robustness–equivariance via contrast sets.** For the two scenarios where we have access to hand-crafted contrast sets, for each model, we plot the robustness of the model on that scenario (worst-case performance across perturbations of each instance) as a function of its standard accuracy.

is the most accurate model for all 9 scenarios. The margin however varies greatly across different scenarios: the largest margin is on **TruthfulQA** where text-davinci-002 achieves accuracy of 62.0% compared to second place from Anthropic-LM v4-s3 (52B) at 35.4%, whereas the smallest margins are for **NaturalQuestions** (closed-book) (38.9% for text-davinci-002 vs. 38.5% for TNLG v2 (530B)) and **HellaSwag** (81.5% for text-davinci-002 vs. 81.1% for Cohere xlarge v20220609 (52.4B)). As this suggests, the most accurate models across the question answering beyond text-davinci-002 is much more variable: all of Anthropic-LM v4-s3 (52B), T0++ (11B), Cohere xlarge v20220609 (52.4B), OPT (175B), TNLG v2 (530B), davinci (175B), and GLM (130B) are a top-3 model in terms of accuracy for some QA scenario, though for 5 scenarios Anthropic-LM v4-s3 (52B) is the second-most accurate and for 6 TNLG v2 (530B) is the third-most accurate.

For calibration, we see several instances of models being very poorly calibrated. For example, on **HellaSwag**, which is one of the scenarios where models have the highest absolute accuracies, the two most accurate models (text-davinci-002, Cohere xlarge v20220609 (52.4B)) both have calibration errors of at least 0.28 (0.286, 0.341 respectively). In fact, we tend to see the J1 models are poorly calibrated on these scenarios, with all 3 models incurring an ECE-10 of roughly 0.5 or more for both **NaturalQuestions** variants, **QuAC**, **NarrativeQA**, and **TruthfulQA**. In contrast, some models are quite calibrated for some scenarios (e.g. Cohere xlarge v20220609 (52.4B) has a calibration error of 0.06 on **NarrativeQA**).

For robustness and fairness, we see all models tend to show consistent drops of 5 to 10 points for all scenarios. Consistent with the broader trends, we find robustness and fairness are strongly correlated with accuracy, with no observed cases of the most accurate models suffering especially large drops for robustness/fairness. One exception is **HellaSwag**, where the three most accurate models are the only models with standard accuracies above 80% (text-davinci-002 = 81.5%, Cohere xlarge v20220609 (52.4B) = 81.1%, Anthropic-LM v4-s3 (52B) = 80.4%), and only text-davinci-002 remains about 70% in the presence of fairness perturbations at 70.3%. Cohere xlarge v20220609 (52.4B) in particular experiences the largest drop of more than 15 points to 66%, which is a worse robust accuracy than TNLG v2 (530B) at 67.8% despite TNLG v2 (530B) being 1.2% worse in standard accuracy. In other words, the model with higher standard accuracy has lower accuracy in the presence of fairness perturbations, with a 3% larger drop due these perturbations. In terms of robustness to semantics-altering perturbations (equivariance), we only have access to Contrast Sets (Gardner et al., 2020) for the **BoolQ** scenario. In Figure 35, we find that the performance of all models drops significantly when evaluated for equivariance. Interestingly, there is a cluster of models with moderate accuracy, but really low robustness: e.g., GPT-J (6B) achieves an accuracy of 55% but a robustness of just 3.6%.

For generative harms, we find the toxicity rates are quite low across all QA scenarios (e.g. less than 1% for all models on both **NaturalQuestions** variant as well as **QuAC**). The one exception is **NarrativeQA**, where several models have toxicity rates between 2–3%, which may be attributable to the story domain

(and/or false positives from the Perspective API being more common for this domain). Looking at biases for **NarrativeQA**, we see models generally show similar biases for all four of {gender, race} × {demographic representation, stereotypical associations}. The one exception is for racial representation, where all but three models achieve bias scores of 0.667 (lower is better/less biased), whereas UL2 (20B) and T5 (11B) have scoress around 0.35 and YaLM (100B) around 0.45. Looking at the model generations, we do see some evidence for discrepancies in model generations, though we also expect the effects referenced are still somewhat weak given models infrequently mention racial words that we track for this scenario. Beyond **NarrativeQA**, models tend to demonstrate similar biases for a given scenario.

**Information retrieval.** To further explore the results for this task, see `https://crfm.stanford.edu/helm/v0.1.0/?group=information_retrieval`. Unless otherwise stated, our results report the scores from the *boosted* setting, which aim in principle to establish an intuitive upper-bound on model quality. Whereas in the *vanilla* settings models re-rank the top-30 passages from BM25, in the boosted setting models re-rank the top-30 passages from BM25 as well as every passage that has an explicit relevance assessment, even when these passages aren't retrieved by BM25. Refer to Section B.2 for a discussion of our IR metrics and a detailed motivation for the boosted setting.

For information retrieval on **MS MARCO (regular)** and **MS MARCO (TREC)**, we begin by observing that the best-performing models show competitive accuracy scores, especially under the boosted setting. The most effective models on **MS MARCO (regular)** are text-davinci-002, which scores 39.8% RR@10 in the boosted setting and 22.4% RR@10 in the vanilla BM25 top-30 setting, and TNLG v2 (530B), which scores 39.7% RR@10 (boosted) setting and 22.5% RR@10 (vanilla), with Cohere xlarge v20220609 (52.4B) being the next most accurate model at 33.1% RR@10 (boosted) and 17.6% (vanilla). To put these scores in perspective, the RR@10 of our underlying classical retriever (i.e., BM25) is 19.0%. That is, in the vanilla top-30 setting, both text-davinci-002 and TNLG v2 (530B) obtain a better ranking than the underlying retriever on average, whereas other models like Cohere xlarge v20220609 (52.4B) perform worse.

We see a similar trend but with larger gains on the more densely-annotated **MS MARCO (TREC)**. In particular, text-davinci-002 is the most accurate model at 65.3% NDCG@10 (boosted) and 61.0% NDCG@10 (vanilla), then TNLG v2 (530B) at 64.7% (boosted) and 60.7% (vanilla), and Cohere xlarge v20220609 (52.4B) at 52.8 (boosted) and 53.4% (vanilla). On **MS MARCO (TREC)**, BM25's NDCG@10 is 50.6%. Given the much more dense annotations in the **MS MARCO (TREC)** setting, we see that these models deliver considerably larger gains over BM25, under both the boosted and the vanilla setting, compared with their gains on **MS MARCO (regular)**. For **MS MARCO (TREC)**, the boosted setting includes all of the assessed passages, including many relevant and many non-relevant passages. Naturally, the gains for the best-performing models are larger in the boosted setting. Perhaps surprisingly, for Cohere xlarge v20220609 (52.4B), quality is diminished in the boosted setting: effectively, this model is distracted by the non-relevant passages that are in the assessed pool more than it benefits from the relevant documents that are not in BM25's top-30 results.

Overall, these few-shot effectiveness scores are relatively competitive but they nonetheless trail the current state-of-the-art retrieval systems by a considerable margin. At the time of writing, in **MS MARCO (regular)**, the top-scoring system on the public leaderboard scores 45% RR@10 on the test set. Compared to this, the models we test score up to 39.8% RR@10 in the boosted setting, which is comparable to recent state-of-the-art standalone retrievers like ColBERTv2 (Santhanam et al., 2021), and 22.5% in the vanilla setting that re-ranks BM25's top-30, which is comparable to simple neural ranker submissions from late 2018 and early 2019 like Conv-KNRM (Dai et al., 2018) and DuetV2 (Mitra & Craswell, 2019). In **MS MARCO (TREC)**, the best-scoring system that participated in the original NIST competition scored 76.5% NDCG@10. Compared to this, the models we test score up to 65.3% NDCG@10 in the boosted setting, which is within 2–3 points of the ANCE (Xiong et al., 2020) system from mid 2020, and 61.0% in the vanilla setting, which is more effective than the Siamese BERT embeddings baseline of Gao et al. (2021c).

These promising accuracy scores for few-shot adaptations of language models are promising, especially since it is not inherently obvious that such models are well-suited for the nuances of these passage ranking scenarios. In particular, these scenarios (at least as realized through our adaptations) necessitate a strong degree of

calibration, in which the probabilities assigned to each of the "Yes"/"No" outputs accurately reflects the continuous degree of relevance of a passage to a query. Out of the MS MARCO corpus with 9M passages, numerous passages can carry varying degrees of relatedness to a given query (Craswell et al., 2020).

Beyond accuracy, we hone in on the robustness and fairness of the systems we study. Robustness and fairness have been considered in many ranking problems (e.g. Zehlike et al., 2017; Singh & Joachims, 2019), but such concerns have only become mainstream in the past few years.[67] We find that text-davinci-002 only presents minor drops in accuracy in the presence of both robustness and fairness perturbations for **MS MARCO (TREC)**, but otherwise most models show some drops, though no model shows especially large drops. To highlight an interesting case, Cohere xlarge v20220609 (52.4B) shows the largest drops for **MS MARCO (regular)**, going from 33.1 standard RR@10 to 29.4 fairness RR@10 and 25.3 RR@10 in the presence of robustness perturbations. We highlight the potential for future work to further explore cross-lingual retrieval settings for both language and dialects, in that we currently perturb both the passages and the query for our dialect perturbations but it may be more realistic to assume the passages are, for example, in SAE but the queries are in AAE to understand performance for AAE speakers.

Finally, in spite of the promising results in terms of model accuracy, we note that scale is critical to many information retrieval applications, necessitating models be efficient. For example, to deploy information retrieval systems on the web in search engines, systems must perform very efficient inference to be useful, which is why amortized indexing strategies have been widely studied and adopted. In contrast, we see these language models when adapted via prompting are drastically less efficient, perhaps also because of the large model sizes (e.g. both of the most accurate models are more than 100B parameters). In particular, the idealized denoised inference time for text-davinci-002 is 0.21s per request (i.e., a query–passage pair to be scored). If these requests have to be issued sequentially per query, this is likely too slow, even though we are evaluating models on a variant where only 30 passages are being ranked for each query. However, it is in principle possible to parallelize such scoring across passages (for a given query), and thus a latency of 0.21s could be acceptable if such parallelism is seen as cost-effective.

**Summarization.** To further explore the results for this task, see `https://crfm.stanford.edu/helm/v0.1.0/?group=summarization`. For summarization on **CNN/DailyMail** and **XSUM**, we begin by observing that the `ROUGE` scores tend to be much lower than our qualitative judgments of summary quality, consistent with Goyal et al. (2022). With that said, broadly, we did find that the `ROUGE-2` scores did correlate with more accurate models across-the-board (e.g. the top models on both datasets based on `ROUGE-2` largely overlapped with those at the top of the accuracy subfigure of Figure 26). In general, we saw a strong correlation with model size, as the largest models tended to be those with high `ROUGE-2` scores for both scenarios, notably with TNLG v2 (530B) having the highest score for both scenarios with a reasonable margin for **XSUM** at 17.9 points when compared with second-place OPT (175B) at 15.6 points and no other model above 14 points.

Beyond model accuracy, we found that getting models to produce summaries of appropriate length (so as to match the distribution of reference summaries in the dataset) was a key challenge, especially given models only observe a few in-context examples and, therefore, may be poorly specialized to the distribution (when compared to models explicitly trained/fine-tuned on the distribution, for which automated `ROUGE` scores may be more useful). In particular, comparing the compression scores across models, we see considerable variation, and the trends were not consistent across the two datasets. As a very striking example, ada (350M) was one of the most compressive on **CNN/DailyMail** but the least compressive on **XSUM** by a wide margin (1.65 vs the next least compressive in babbage (1.3B) at 6.12 where higher scores means more compression), suggesting the ada (350M) model especially did not "understand" the specification of length requirements in the instructions in the prompt. In terms of the degree of abstraction in model generations, we found that the relationship between model quality and abstraction (measured in terms of both coverage and density from Grusky et al. (2018)) was very variable, with few consistent trends.

Since the summarization scenarios required the longest-form generation of all core scenarios, we paid special attention to the presence of generative harms. For stereotypical associations (for both race and gender) on both datasets, we found that all models exhibited very similar biases, especially on **CNN/DailyMail**. In

[67]For example, see the TREC Fair Ranking track: `https://fair-trec.github.io/`.